

Counting White-Tailed Deer in Vehicle-Mounted Thermal Video

How the pipeline works, what it produces, and where it loses animals

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One annotated frame from the corpus: 640 x 512 px thermal, contrast-normalized

A survey needs animals counted, not boxes placed

What a detector score asks

Is each box in the right place, frame by frame? Scored with mean average precision (mAP).

What a survey asks

How many different animals were on this transect? Each animal must be counted once.

Between the two sit three decisions that no detection score measures:

- linking the boxes of one animal over time
- deciding which tracks are real animals
- deciding which tracks repeat an animal already counted

Our question: does better detection give a better count?



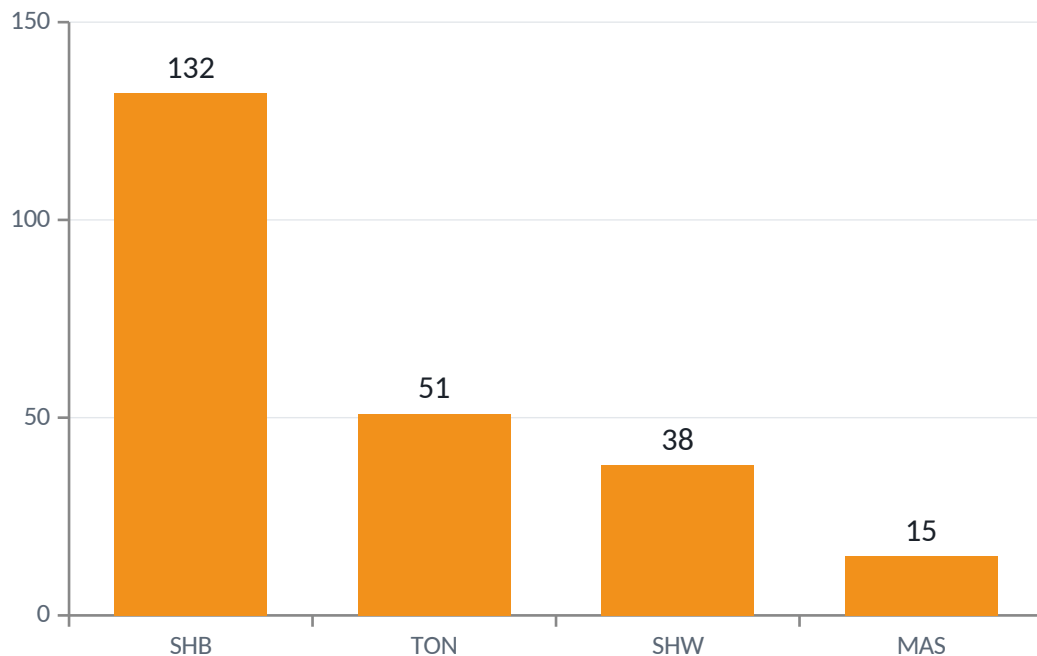
What the computer receives



Annotation: one box per deer, the same identity in every frame

The data: 32 transects, every deer followed by hand

Annotated deer per site (236 in total)



32

videos, 8 per site, recorded at night from the vehicle

521,930

frames, 640 x 512 px at 60 frames per second

236

individual deer, each one a track from first to last frame

29 x 24 px

median size of a deer in the image

Training

19 videos, 153 deer

The detector learns from these.

Validation

4 videos, 45 deer

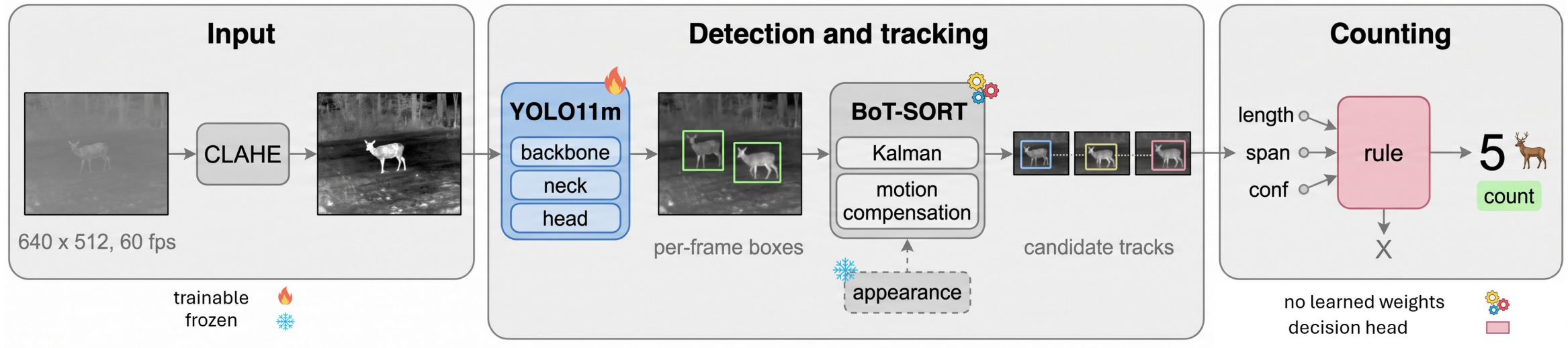
Used only to pick the best detector checkpoint.

Test

9 videos, 38 deer

Never seen during training.

The pipeline in one picture



1 Contrast

Make faint animals visible (CLAHE).

2 Detection

Box and confidence for each deer, in each frame.

3 Tracking

Link the boxes of one animal over time.

4 Confirmation

Accept a track or reject it. Count the accepted tracks.

Deterministic: the same video always gives the same boxes, tracks and count (three repeated runs were identical).

Contrast normalization makes faint animals visible



As recorded by the camera



After contrast normalization (CLAHE)

What it does

Stretches contrast inside small tiles of the image (8 x 8 grid), with a limit so that noise is not amplified.

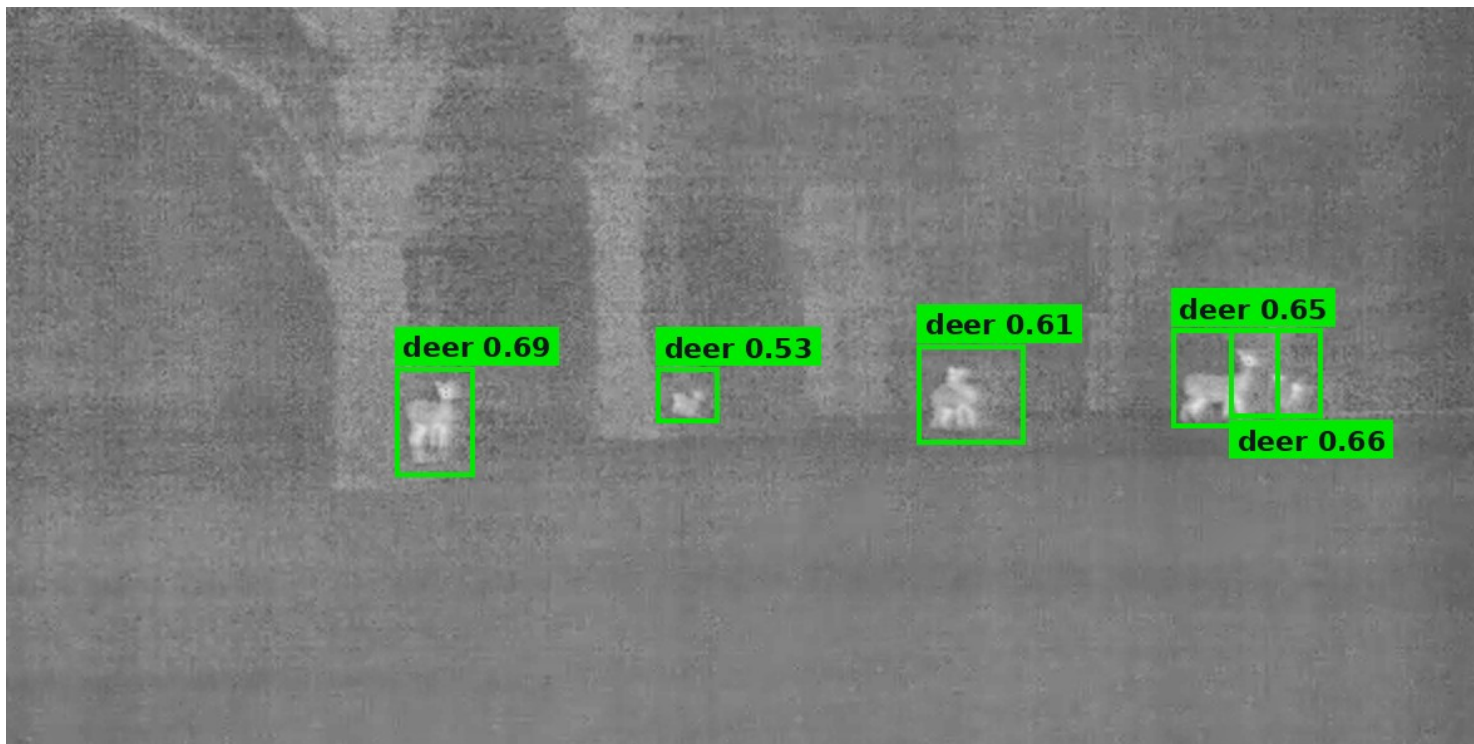
Why we need it

Thermal contrast is low and it drifts during a transect. A deer is only slightly warmer than the ground behind it.

0.30 to 0.52

detection score (mAP50) on test videos, without and with normalization. Larger than any difference between models.

The detector finds deer in each frame, one frame at a time



Detector output on one frame of a held-out video (GolfDr, SHB). Numbers are the confidence of each box.

- Input: one contrast-normalized frame.
- Output: a box and a confidence (0 to 1) for each deer it finds.
- No memory: every frame is processed alone.
- Small animals give modest confidence, so all boxes above 0.10 are passed on.

YOLO11m

detector, 640 px input

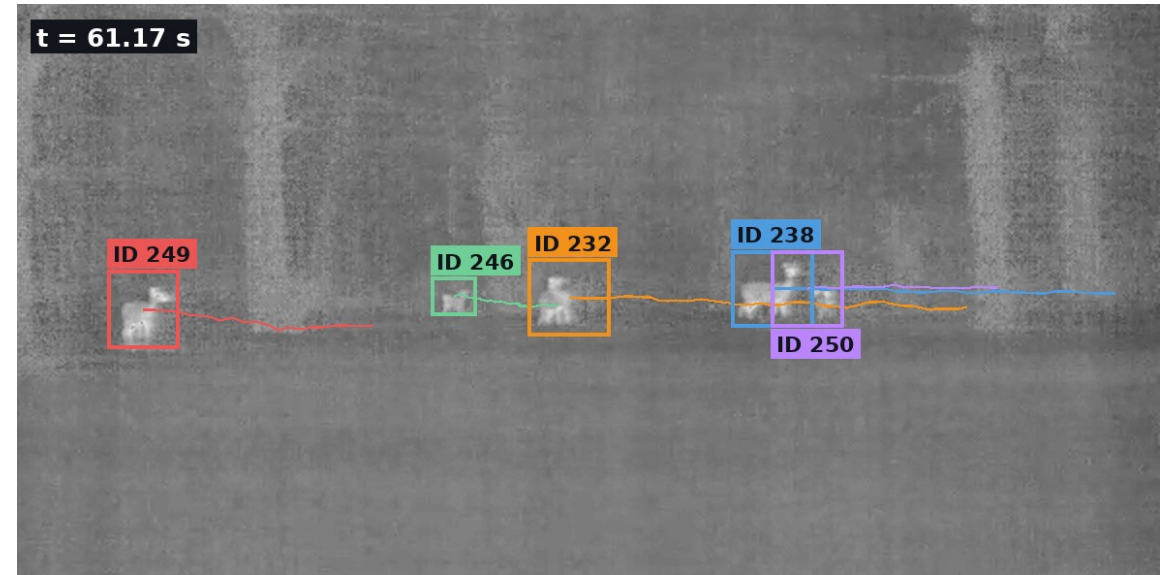
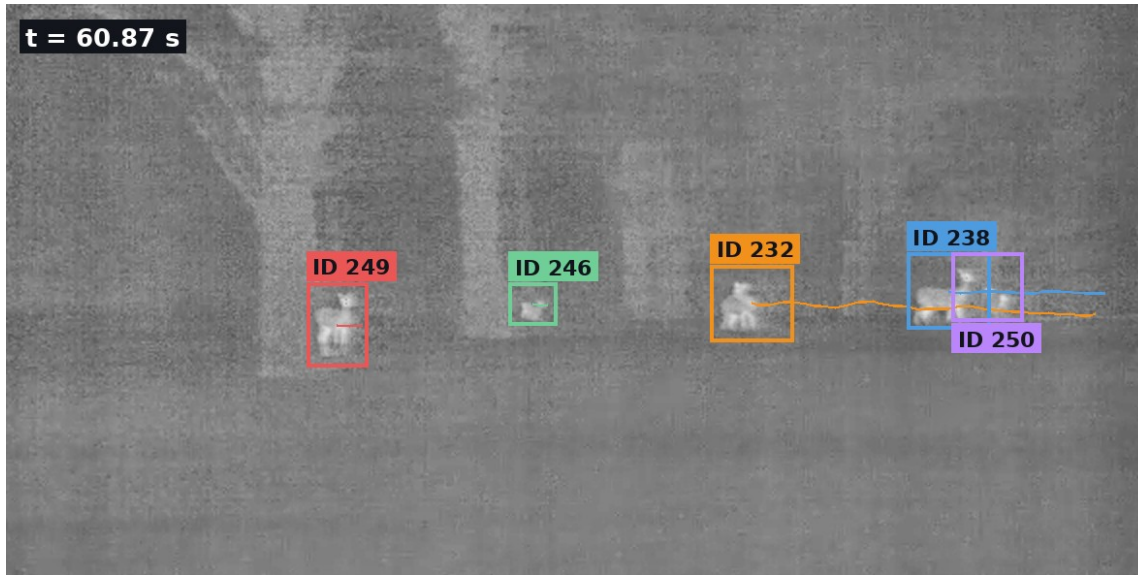
12,852 frames

for training, 7,839 of them with no deer

11 models

compared on the same data (slide 12)

The tracker links boxes into one track per animal



Same scene 0.3 s apart. Colour and ID stay with the animal while the camera moves; lines are the path of each track.

Predict

A Kalman filter predicts where each animal will be in the next frame.

Remove camera motion

Optical flow estimates how the vehicle moved the image, and that shift is removed.

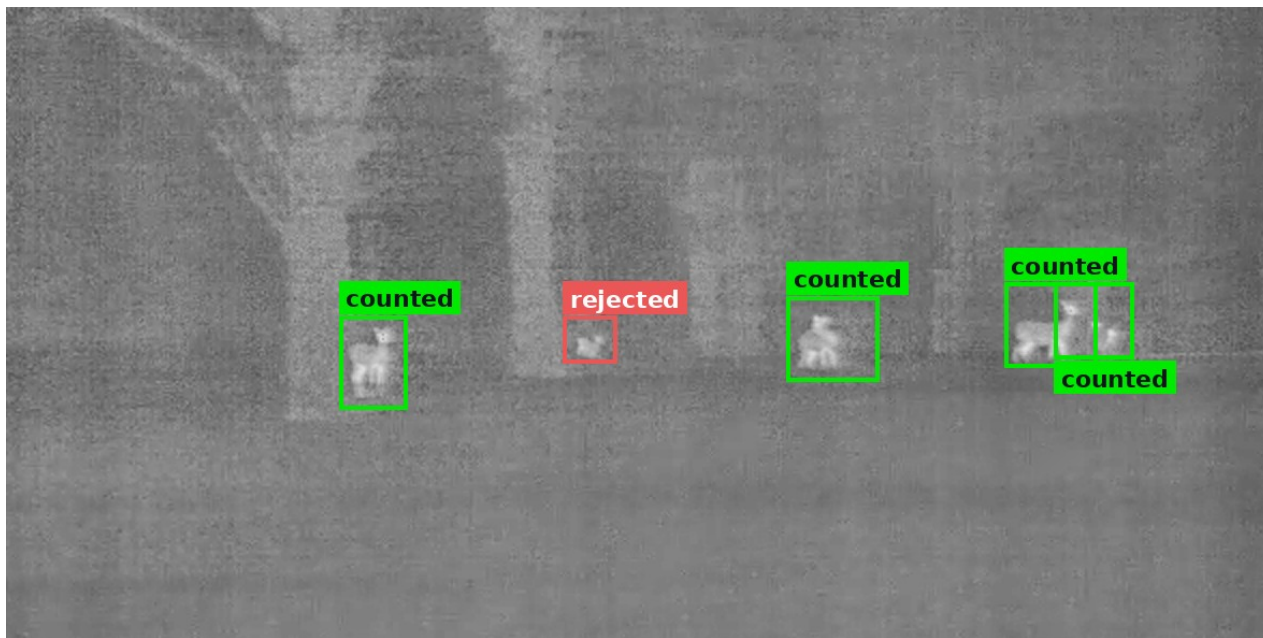
Match

New boxes are matched to predicted positions by overlap. Unmatched boxes are recovered if they line up in time (17 animals).

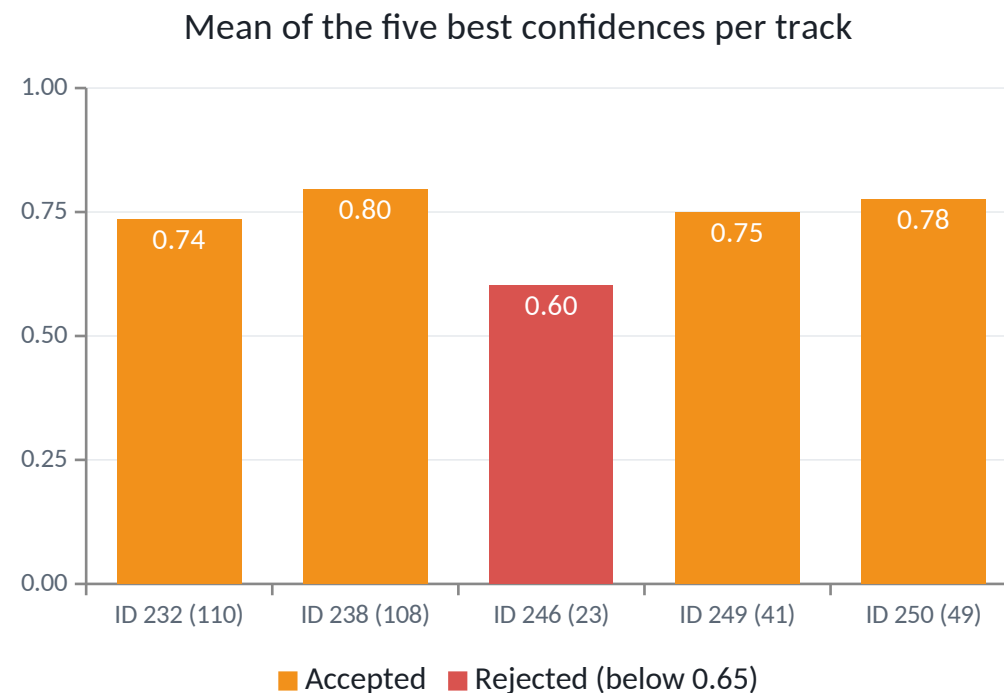
Not identification

Position and motion only. The system does not recognise individual deer by appearance.

A simple rule decides which tracks are counted



Decision of the rule for the five tracks in this scene. The rejected animal is a real, annotated deer.



Condition 1: length

At least 20 detections, which is one third of a second of video.

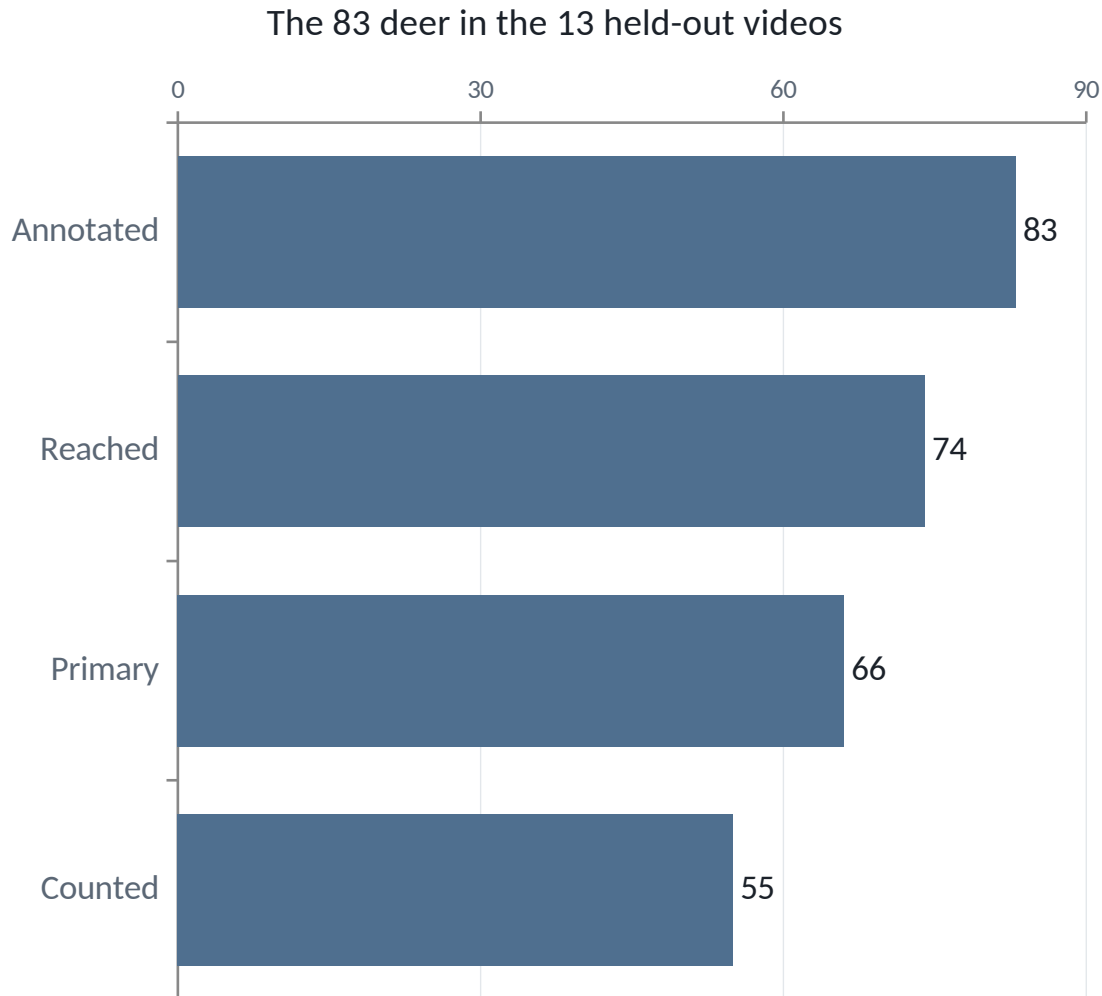
Condition 2: confidence

The five best detections of the track must average 0.65 or more.

Not set by hand

Chosen by a search over 378 settings on the training videos, then frozen.

How we measure where animals are lost



1 REACHED

If not: lost by detection or tracking

Did any candidate track touch the deer in at least one frame?

2 PRIMARY

If not: lost by association

Does the deer have a track of its own, not shared with another deer?

3 COUNTED

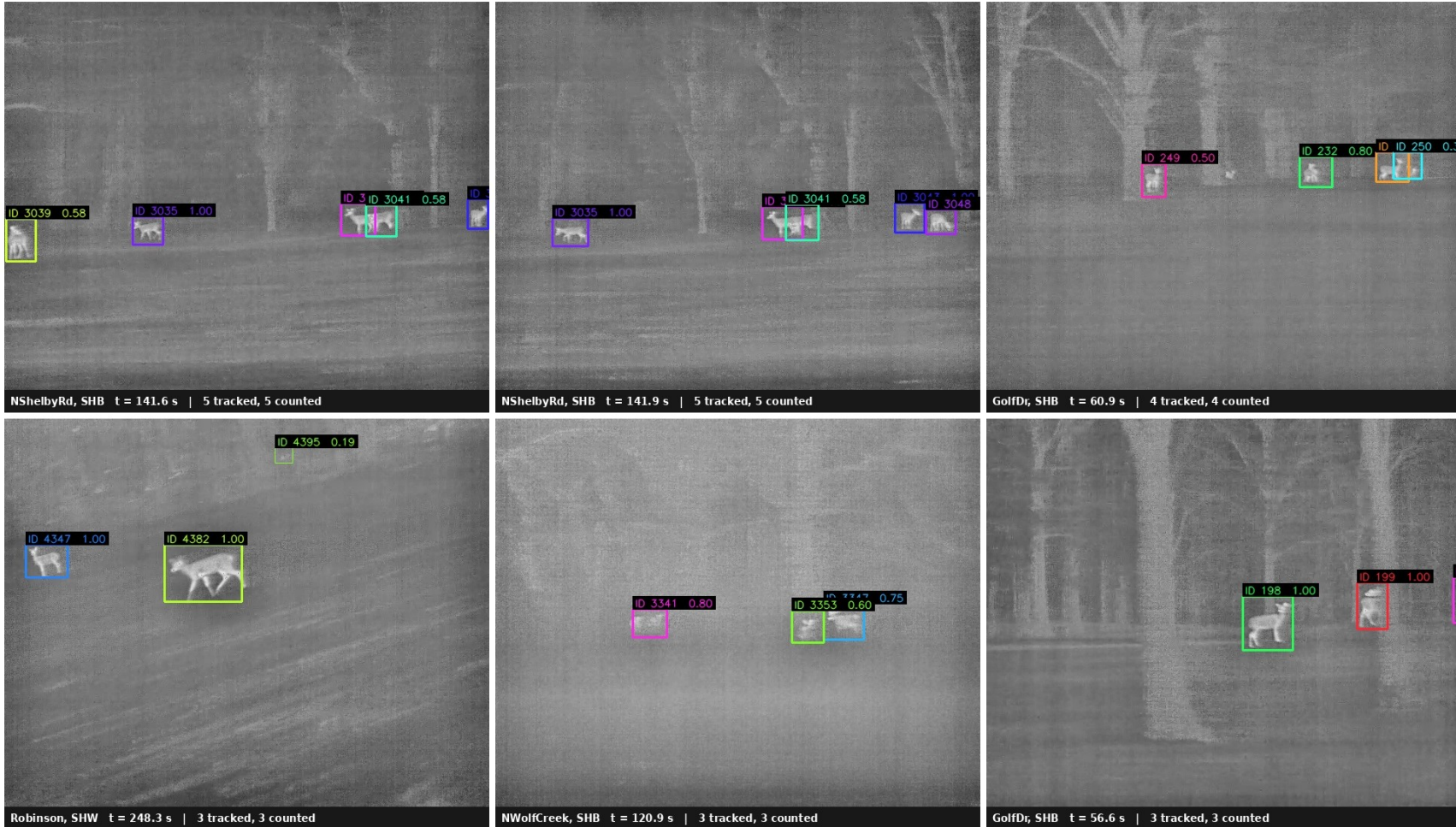
If not: lost by confirmation

Did the rule accept a track on this deer?

Held-out videos only

Results come from 13 videos the detector never trained on. Measured on all 32 videos, coverage looks 15.5 points higher.

Output: tracks through overlap and low contrast



Identity persists

Top left pair: same scene, 0.3 s apart, five deer, five tracks.

Overlap

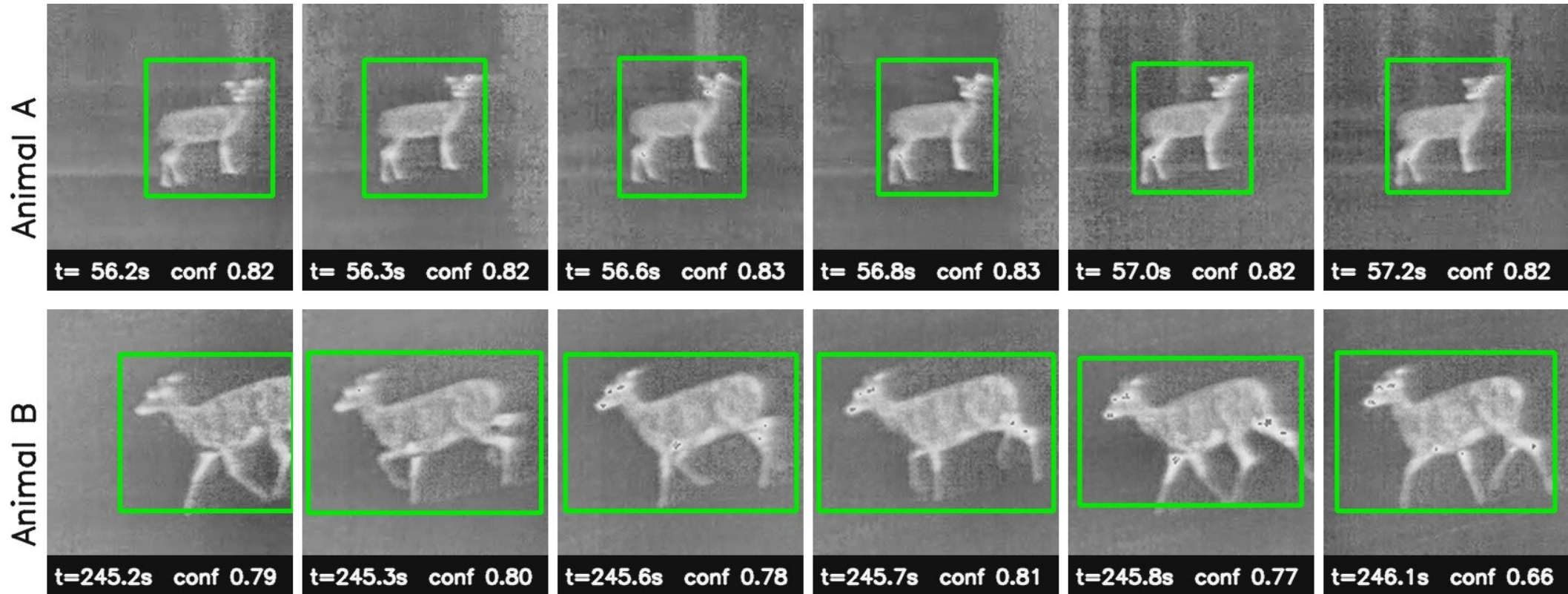
Deer standing together are kept as separate tracks.

Low contrast

Two faint deer on a uniform background are still two tracks.

Frames from four held-out transects (NShelbyRd, GolfDr, Robinson, NWolfCreek).

Output: evidence and a confidence for every counted deer



92.1%

of counted deer have confidence 0.80 or higher

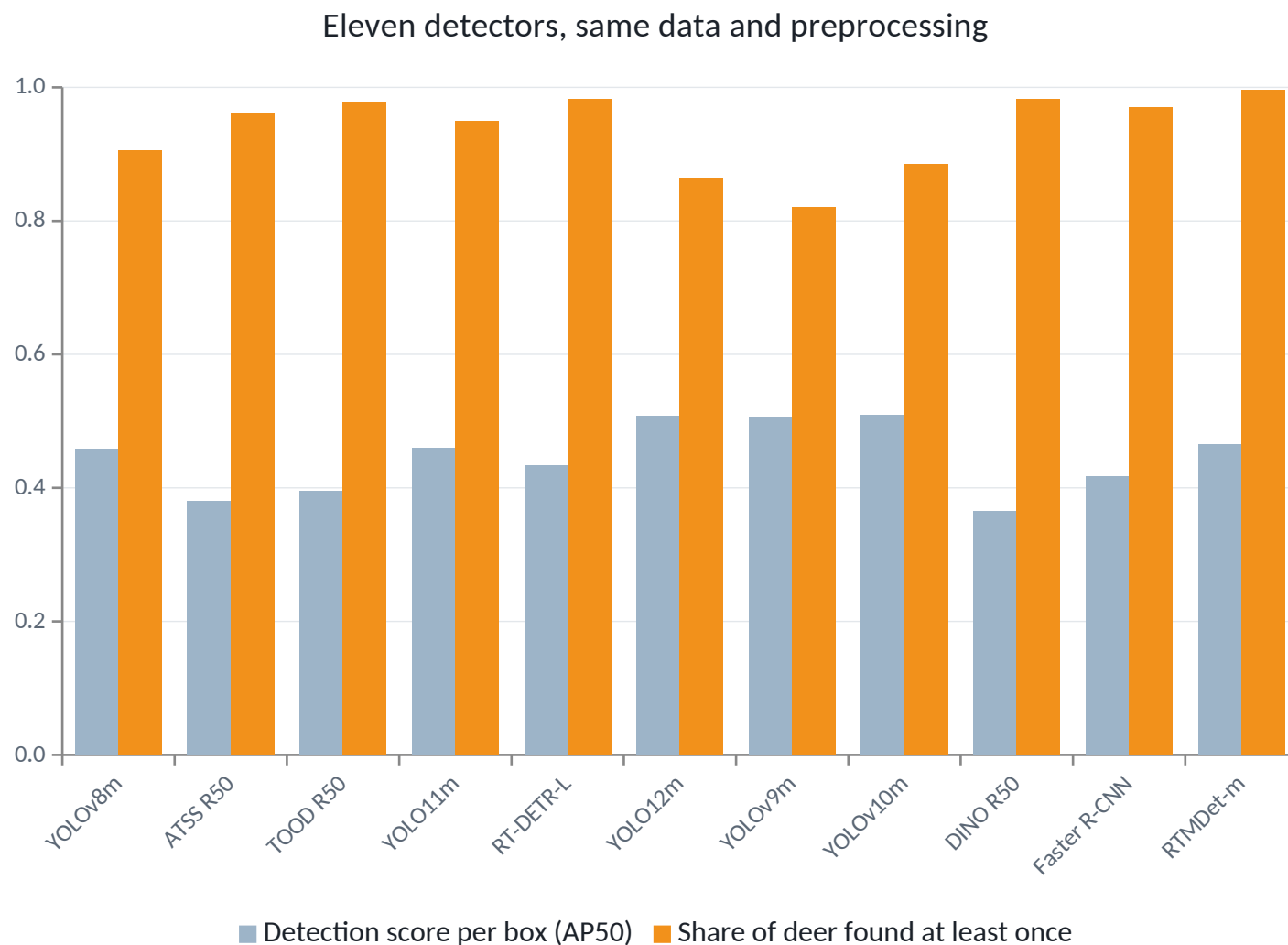
793 tracks

with low confidence are flagged for human review

One sheet per deer

so every count can be checked by eye

Result 1: detection is close to solved



82 to 99.6%

of deer are found at least once by every detector

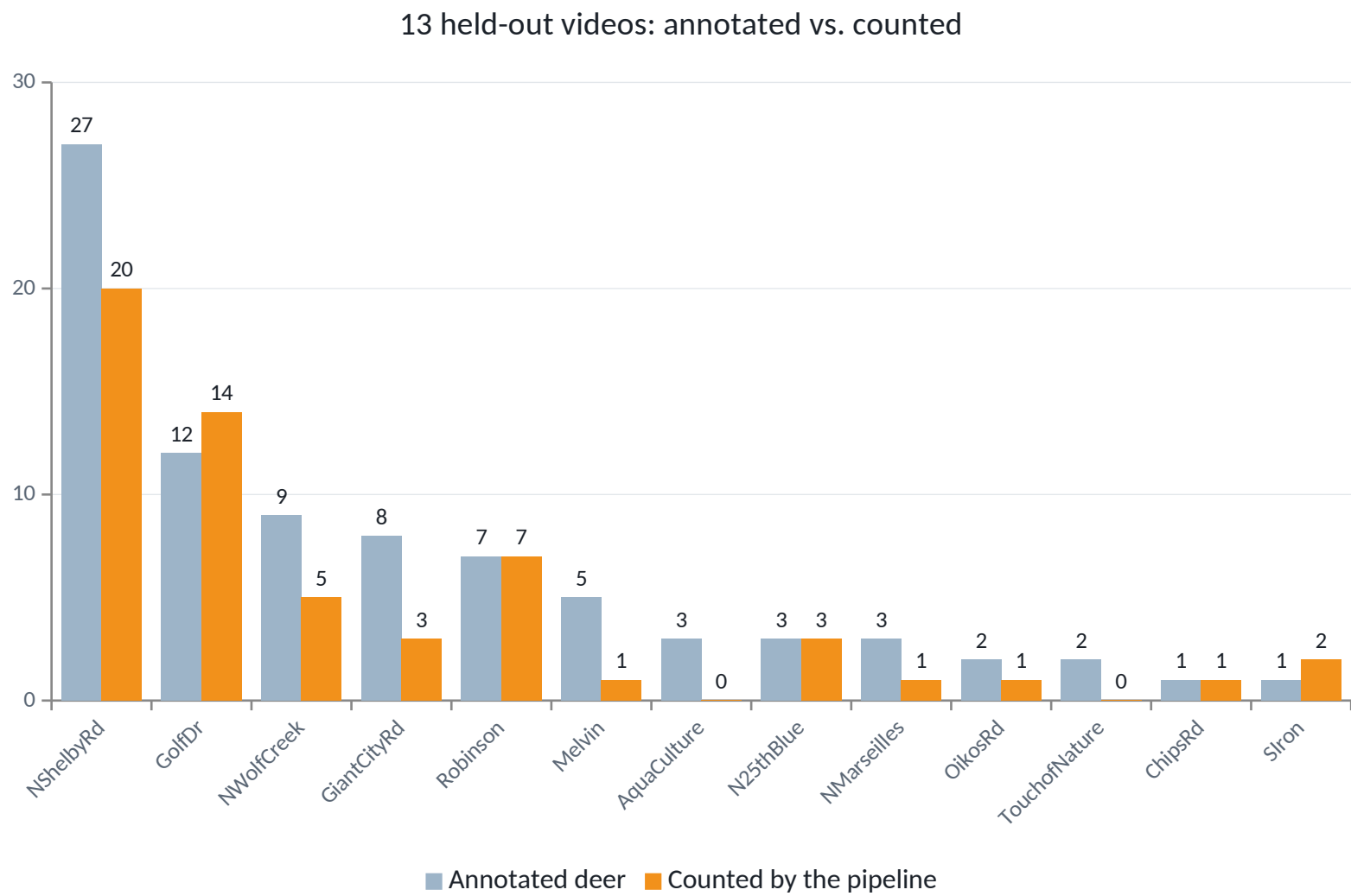
0.37 to 0.51

is the usual per-box score for the same models. It is strict for 27-pixel animals.

Our choice: YOLO11m

Chosen for few false alarms (precision 0.90), not for the best per-box score.

Result 2: counts per transect on unseen videos



58 of 83

deer reported over the 13 videos

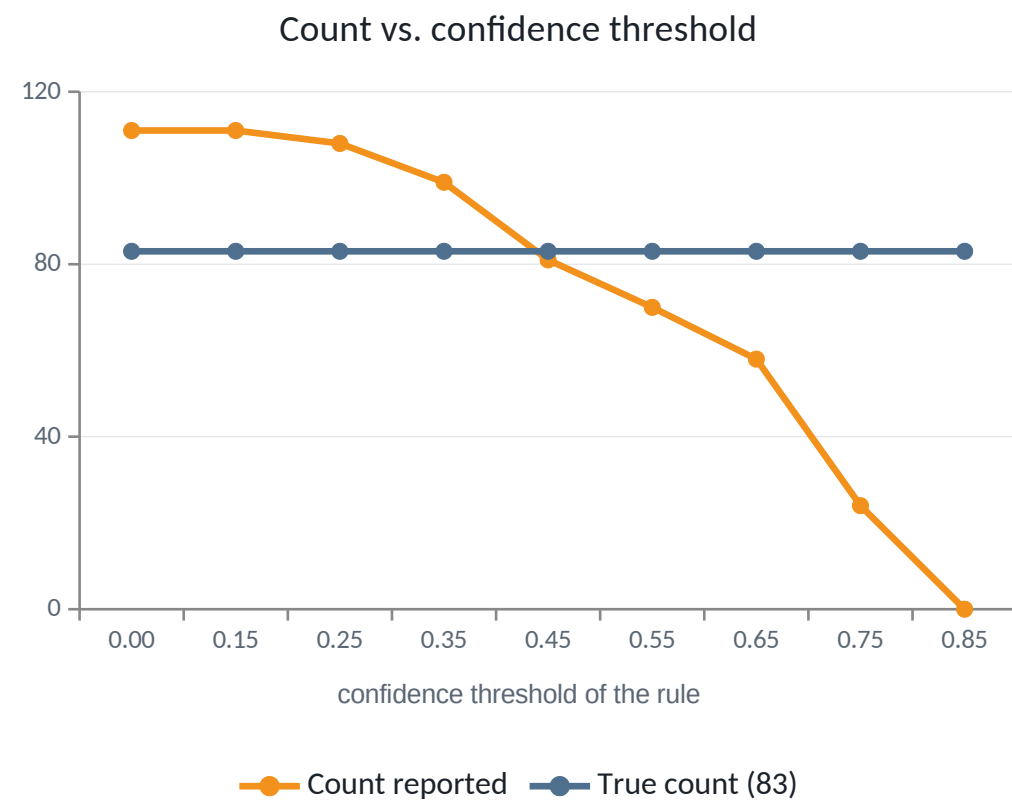
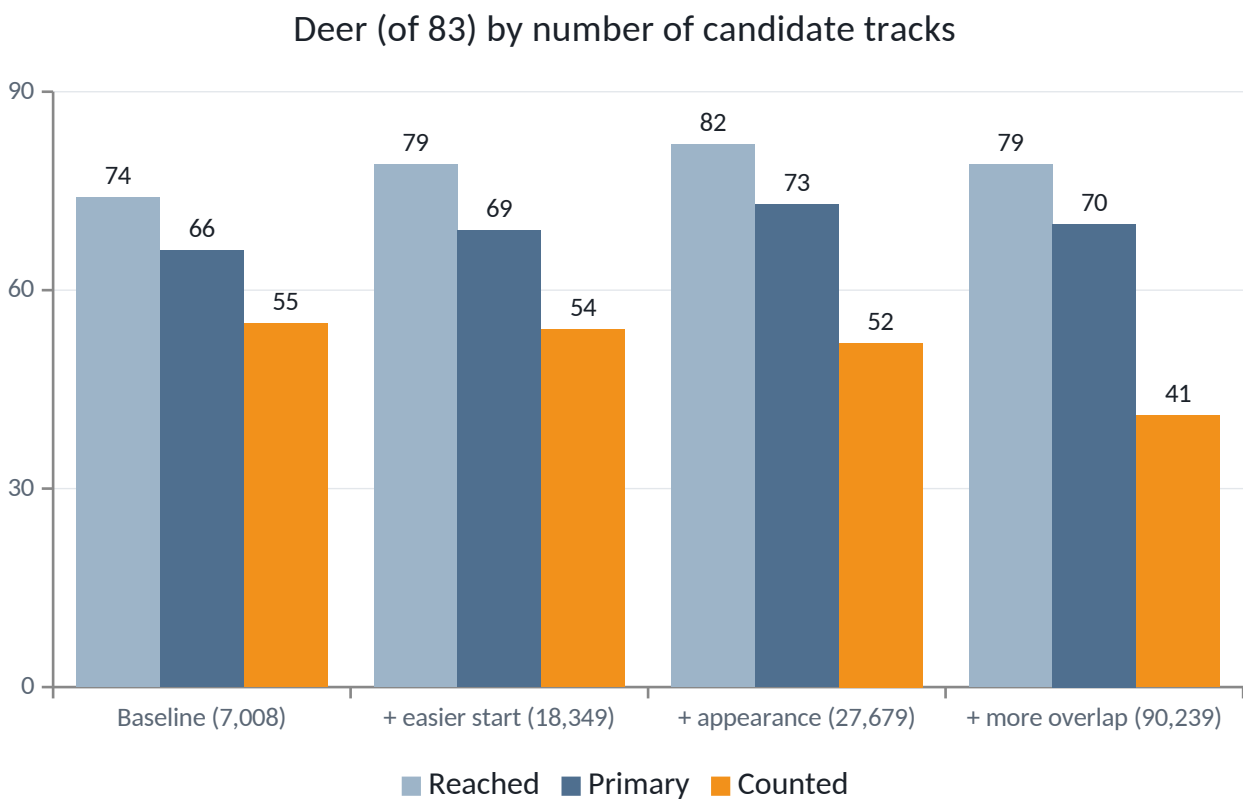
2.38

mean absolute error, in deer per video

Under-counts

Only 2 of 13 videos are counted too high.
The error is in the safe direction.

Result 3: more candidates, fewer deer counted

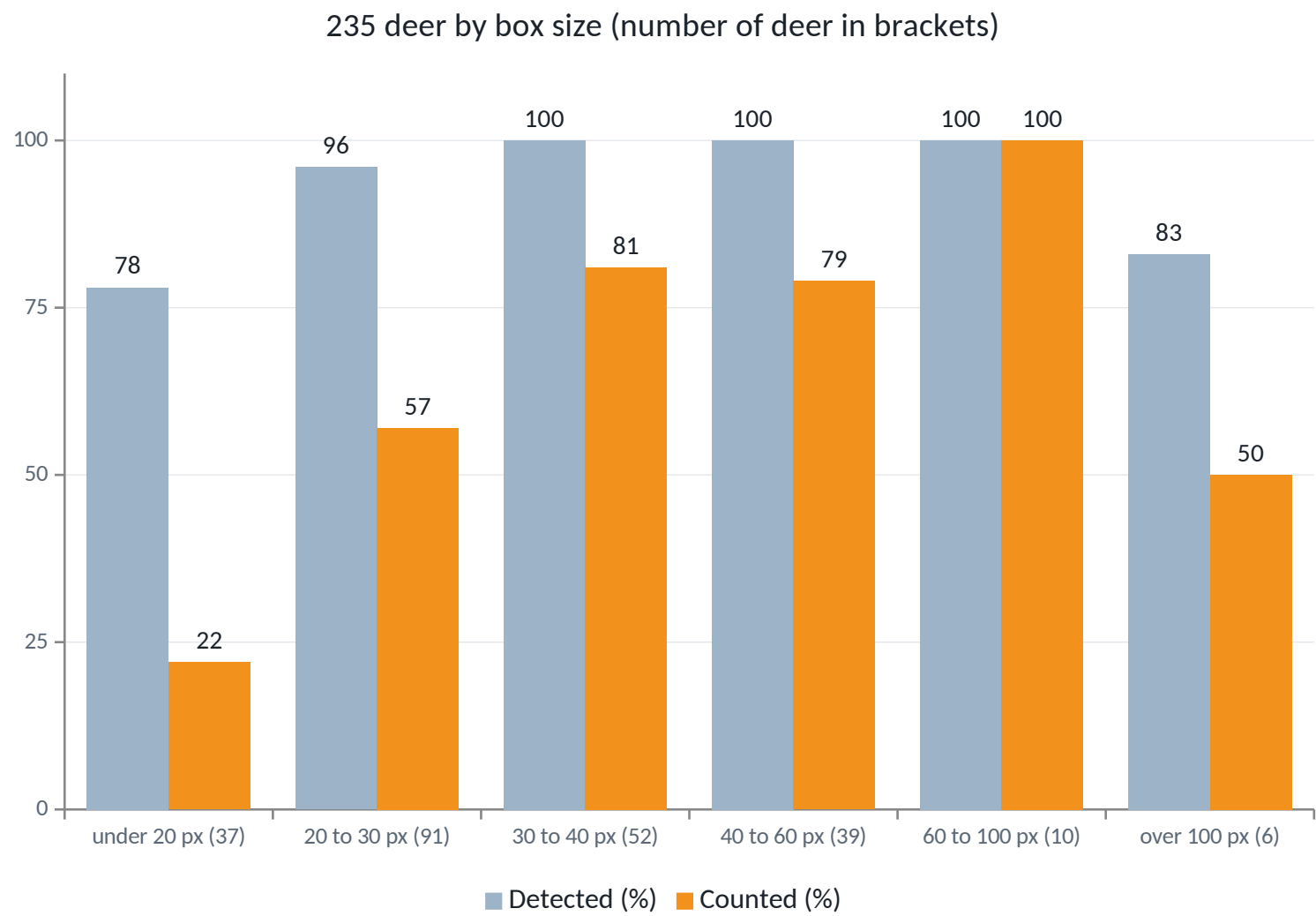


More candidates reach more deer but count fewer. The extra tracks are second sightings of deer already counted.

Lowering the threshold brings back missed deer, with about two duplicates for each one.

Machine learning in place of the rule was worse: error 2.85 to 3.00 deer per video, against 2.38.

Result 4: chance of being counted, by size in the image



Distance costs counts

Small, distant deer are usually detected, but their tracks are too weak to be accepted.

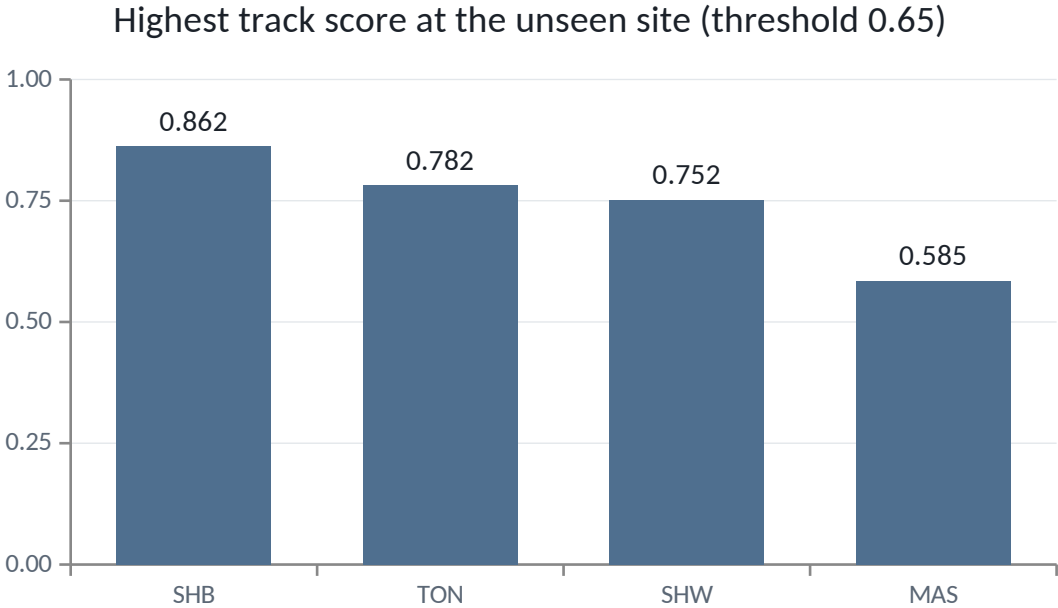
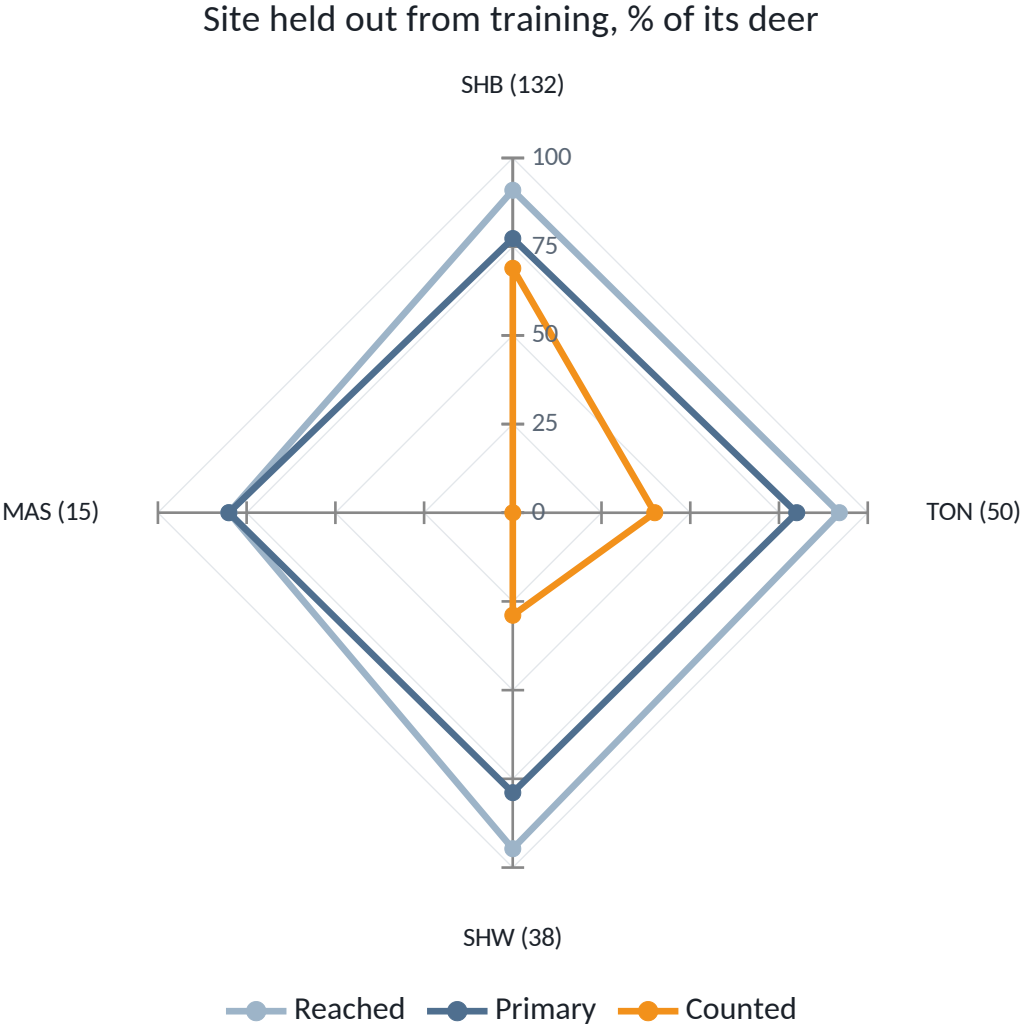
Box size is a proxy

It mixes distance with body size and posture. It needs checking against real distances.

A detection function?

This curve could feed distance sampling. The annotation itself thins out below about 20 px.

Result 5: a new site needs its own threshold



What transfers

Detection and tracking: 91% reached and 78% primary over the four sites.

What does not

The fixed threshold. One night per site, so site and weather cannot be separated.

What we learned, and what we would like to do with you

What we learned

1

Detection is not the bottleneck

Per animal, almost every deer is found. Confirmation is where deer are lost.

2

More candidates did not help

Better detection, tracking and looser filtering each left the count equal or lower.

3

The limit is data, not the model

About 200 real tracks. Duplicates look like real deer, because they are real deer.

4

The threshold is a choice

0.65 for one transect; 0.45 for an almost unbiased total (81 of 83).

What we would like to do with you

Box size against real distance

Validate the proxy, then build a detection function for distance sampling.

Repeated nights at one site

To separate the effect of the site from the weather of the night.

A small annotated sample per new site

To set the threshold again before counting there.

Dataset, code and a software tool

Video in; video with boxes, tracks and the count out.